COMPLETE NETWORKING EXAMINATION NOTES

Cisco Networking • TCP/IP • OSI • Routing • Switching • Security

60 MCQs | 2 Hours

Easy-to-learn, syllabus-based study material with examples, command references, revision tables and practice MCQs.

How to study

- Study one chapter at a time. Learn the comparison tables and bold terms first.
- Practise subnetting calculations on paper rather than only reading solutions.
- Attempt MCQs before looking at the answer and explanation.
- Use the final revision sheets for quick recall before the examination.

Complete syllabus checklist

Syllabus section	Coverage in this PDF
Network basics	Characteristics, network types, topologies, devices, network architecture, communication types and data communication elements.
Cisco administration	Packet Tracer, CLI modes, help, hostname, passwords, banners and saving configurations.
OSI/TCP-IP	Seven OSI layers, TCP/IP layers, encapsulation, de-encapsulation, peer communication and layer services.
Physical/Data Link/Network	Copper/fibre, straight-through/crossover, signalling, MAC, Ethernet, IP, routing and packet forwarding.
Transport/Application	TCP, UDP, ICMP, FTP, TFTP, HTTP/HTTPS, POP3, IMAP, SMTP, DHCP and DNS.
IP addressing	IPv4, IPv6, classes, public/private, network/host/broadcast, FLSM and VLSM.
Routing	Static, RIP, EIGRP, OSPF single area and redistribution.
Administration/security	Telnet, SSH, TFTP backup/restore, password recovery, standard/extended ACL, named/numbered ACL and NAC.
NAT	Static NAT, Dynamic NAT, DNAT and PAT.
Switching	VLAN, DTP, VTP, STP, port security, Router-on-a-Stick and SVI.
Additional theory	Firewall, proxy and web services.

EXAM ALERT: The syllabus images are the authority. Topics repeated across the two images are combined to avoid repetition, but all named concepts and labs are retained. ICMP is listed under Transport in one image; technically it is an Internet/Network-layer control protocol, not TCP/UDP.

1. Network fundamentals

1.1 Definition and purpose

A computer network is two or more computing devices connected through wired or wireless media so they can exchange information and share resources. Networks support file/printer sharing, communication, internet access, centralized management, backup, collaboration and remote access.

• Sender: device that transmits data.

• Receiver: device that accepts data.

• Message: information being sent.

• Medium: path such as copper, fibre or radio.

• Protocol: agreed rules for format, timing and meaning.

1.2 Network characteristics

Characteristic	Meaning / example
Fault tolerance	Continues operating after a failure; uses redundant links, devices or paths.
Scalability	Can grow in users, devices, traffic and services without unacceptable performance loss.
QoS	Prioritizes and manages traffic to meet application needs, e.g. voice/video.
Security	Protects confidentiality, integrity and availability through authentication, authorization, encryption and filtering.

Performance terms: Bandwidth is capacity; throughput is actual achieved rate; latency is delay; jitter is variation in delay; packet loss is the fraction of packets that do not arrive. Real-time voice/video is sensitive to latency, jitter and loss.

1.3 Network types

Type	Coverage / example
PAN	Personal area; Bluetooth phone, watch and headset.
LAN	Room/building/campus; home or office Ethernet/Wi-Fi.
CAN	Campus area; university buildings connected together.
MAN	Metropolitan/city area; branches across a city.
WAN	Country/continent/world; bank branches, Internet.
SAN	Dedicated storage network connecting servers to shared storage.
VPN	Logical private tunnel over an existing network; not a geographic size category.

EXAM ALERT: PAN is generally smallest and WAN largest. CAN can mean Campus Area Network; SAN usually means Storage Area Network in storage contexts.

1.4 Network architecture

Client-server: a client requests a service and a server provides it. Centralized management, security and backup are advantages. Server cost, bottlenecks and server dependency are disadvantages.

Peer-to-peer: peers can both request and provide resources without requiring a dedicated central server. It is simple and economical for small groups, but centralized security, backup and administration are harder.

Feature	Client-server	Peer-to-peer
Control	Centralized	Distributed
Dedicated server	Usually required	Not required
Management	Easier to enforce centrally	Spread across peers
Typical fit	Enterprise services	Small/simple sharing

Convergence means carrying voice, video and data services over shared network infrastructure. Architecture design goals include performance, availability, scalability, security and manageability.

1.5 Network topologies

Topology	Structure	Key strength / weakness
Bus	All devices share one backbone.	Simple; backbone failure affects all; terminators at ends in legacy coax networks.
Star	Each device connects to central hub/switch.	Easy troubleshooting; central-device failure affects the star.
Ring	Each node connects to two neighbours in a loop.	Token-based designs can provide predictable access; basic ring can fail if a link breaks.
Full mesh	Every node directly links to every other node.	High redundancy, high cost.
Partial mesh	Only selected nodes have multiple links.	Balances redundancy and cost.
Tree	Hierarchical branches, often multiple stars.	Scalable hierarchy; upper-level failures affect branches.
Hybrid	Combination of topologies.	Flexible but complex.

Full-mesh links: $L = n(n-1)/2$. Five nodes need $5\times4/2 = 10$ links; six need 15.

1.6 Communication modes

Term	Meaning	Example
Simplex	One-way only.	Broadcast transmission.
Half duplex	Both directions, not simultaneously.	Walkie-talkie.
Full duplex	Both directions simultaneously.	Modern switched Ethernet.
Unicast	One sender to one receiver.	Web client to server.
Broadcast	One sender to all devices in a broadcast domain.	IPv4 ARP request.
Multicast	One sender to a subscribed group.	IP multicast stream.
Anycast	Same service address at multiple locations; routing selects one.	Distributed DNS/CDN.

IPv4 multicast is 224.0.0.0–239.255.255.255 (224.0.0.0/4). Limited broadcast is 255.255.255.255. Routers do not forward ordinary Layer 2 broadcasts by default.

Practice MCQs

1. Which characteristic helps a network continue after a link failure?
A. Scalability
B. Fault tolerance
C. Jitter
D. Convergence

Answer: B — Redundant paths provide fault tolerance.

2. Which network type generally spans a city?
A. PAN
B. LAN
C. MAN
D. WAN

Answer: C — MAN is metropolitan-area networking.

3. A full mesh of 6 nodes requires how many links?
A. 6
B. 12
C. 15
D. 30

Answer: C — $n(n–1)/2 = 6×5/2 = 15$.

4. One sender to a selected group is:
A. Broadcast
B. Multicast
C. Unicast
D. Simplex

Answer: B — Multicast targets group members.

2. Network devices

Device	Main role	Layer / key fact
NIC	Connects host to network; sends/receives frames.	Physical and Data Link functions; MAC address.
Hub	Repeats incoming signal to other ports.	Layer 1; no MAC learning; one shared collision domain.
Repeater	Regenerates/retimes signals to extend reach.	Layer 1; does not filter by MAC/IP.
Switch	Learns source MAC and forwards Ethernet frames.	Layer 2; each port is its own collision domain.
Router	Forwards IP packets between networks.	Layer 3; uses routing table; separates broadcast domains by default.
Modem	Modulator-demodulator; adapts signals to access medium.	Access technology dependent.
Load balancer	Distributes client requests among servers.	May use Layer 4 or Layer 7 information.
Firewall	Allows/blocks traffic based on security policy.	May inspect Layers 3–7.

2.1 Switch operation

A switch learns the **source MAC address** of each incoming frame and associates it with the ingress port in its MAC/CAM table. It then looks up the destination MAC. A known unicast is forwarded to the relevant port. An unknown unicast is flooded to eligible ports in the same VLAN except the ingress port. Broadcast frames are flooded within the VLAN.

2.2 Hub, switch and router comparison

Feature	Hub	Layer 2 switch	Router
Layer	1	2	3
Decision	No address learning	Destination MAC	Destination IP/routing table
Function	Repeats signals	Forwards frames	Forwards packets between networks
Collision domains	One shared	One per port	Per interface segment
Broadcast domains	Does not divide shared LAN	VLANs define domains	Separates by default

A modem's name comes from **modulator + demodulator**. A NIC may be wired or wireless; one computer may have several NICs. A load balancer uses round robin, weighted round robin, least connections or IP hash. Firewalls may be stateless packet filters, stateful firewalls, proxies/application firewalls or next-generation firewalls.

Practice MCQs

1. Which device learns source MAC-to-port mappings?
A. Hub
B. Switch
C. Repeater
D. Modem

Answer: B — A switch builds a MAC address table.

2. Which device forwards between IP networks?
A. Hub
B. NIC
C. Router
D. Repeater

Answer: C — Routers make Layer 3 forwarding decisions.

3. An unknown unicast frame is normally flooded:
A. Across all VLANs
B. Within the same VLAN except ingress
C. Only to the router
D. Nowhere

Answer: B — Unknown-unicast flooding is VLAN scoped.

3. OSI and TCP/IP models

OSI layer	Name	Function / examples
7	Application	Services used by applications: HTTP, FTP, DNS.
6	Presentation	Translation, encoding, compression and encryption concepts.
5	Session	Establishes, manages and terminates dialogs/sessions.
4	Transport	End-to-end delivery, segmentation, reliability, ports: TCP/UDP.
3	Network	Logical addressing and routing: IP; packet.
2	Data Link	Framing, MAC addressing, media access, error detection: Ethernet; frame.
1	Physical	Bits, signals, cables, connectors and radio.

EXAM ALERT: Mnemonic from Layer 7 down: All People Seem To Need Data Processing. From Layer 1 up: Please Do Not Throw Sausage Pizza Away.

TCP/IP model (4 layers)	OSI mapping
Application	OSI 5–7
Transport	OSI 4
Internet	OSI 3
Network Access / Link	OSI 1–2

Some teaching material uses a five-layer Internet model by separating Data Link and Physical. The four-layer and five-layer representations are both common.

3.1 Encapsulation and de-encapsulation

```
Application data
↓ Transport adds header → segment (TCP) / datagram (UDP)
↓ Network adds IP header → packet
↓ Data Link adds frame header/trailer → frame
↓ Physical transmits bits
Receiver reverses the process: de-encapsulation.
```

Each layer adds protocol control information as data moves down the sender's stack. The receiver removes/interprets it as data moves up. Peer-to-peer communication means corresponding protocol layers logically communicate with their counterparts.

Layer	PDU
Application / Presentation / Session	Data
Transport	TCP segment / UDP datagram
Network	Packet
Data Link	Frame
Physical	Bits

Practice MCQs

1. Which layer performs routing?
A. Data Link
B. Network
C. Session
D. Presentation

Answer: B — OSI Layer 3 routes packets.

2. Which layer uses TCP/UDP port numbers?
A. Physical
B. Network

C. Transport
D. Data Link

Answer: C — Ports identify application endpoints.

3. A Data Link PDU is a:
A. Bit
B. Frame
C. Packet
D. Segment

Answer: B — Layer 2 creates frames.

4. Physical layer, media and cables

The Physical layer transmits raw bits as electrical, optical or radio signals. It defines media, connectors, pin layouts, signalling, timing, data rates and physical transmission characteristics.

Medium	How it works	Advantages / limitations
UTP copper	Twisted copper pairs carry electrical signals.	Affordable and common; twisting reduces interference; susceptible to EMI compared with fibre.
STP copper	Shielded twisted pairs.	More protection from interference; costlier and needs correct grounding.
Coaxial	Central conductor with insulation and shield.	Good shielding; cable broadband and legacy Ethernet.
Fibre optic	Light pulses through glass/plastic fibre.	High bandwidth, long distance, immune to EMI; optics/termination can cost more.

Single-mode fibre has a small core and is suited to longer distances. Multimode fibre has a larger core and is commonly used for shorter links. Attenuation is signal weakening; noise is unwanted energy; interference is disturbance from other signals.

4.1 Straight-through versus crossover

Cable	Traditional use	Wiring
Straight-through	Unlike devices: PC-to-switch, router-to-switch.	Same standard at both ends: T568A-A or T568B-B.
Crossover	Like devices in older Ethernet: switch-switch, PC-PC, router-router Ethernet.	Transmit/receive pairs crossed; commonly T568A at one end and T568B at the other for 10/100BASE-TX.

EXAM ALERT: Modern Auto-MDI/MDIX equipment can automatically adapt, so either cable may work. MCQs often ask the traditional rule.

Practice MCQs

1. Which medium carries data as light?
A. UTP
B. Coaxial
C. Fibre
D. STP

Answer: C — Fibre uses optical signals.

2. Attenuation is:
A. Signal weakening
B. Address translation
C. Frame flooding
D. Encryption

Answer: A — Signal strength decreases over distance.

5. Data Link layer, Ethernet and MAC

• Framing: wraps network packets in frames.

• MAC addressing: identifies interfaces on a local link.

• Media access: coordinates access to a shared medium where required.

• Error detection: Ethernet FCS/CRC detects corruption; standard Ethernet does not guarantee correction.

• LLC (Logical Link Control) provides a logical interface to Layer 3; MAC handles addressing and media access.

A traditional Ethernet MAC address is 48 bits (6 bytes), written in hexadecimal, e.g. 00:1A:2B:3C:4D:5E. The first 24 bits commonly identify the vendor OUI in universally administered addresses. FF:FF:FF:FF:FF:FF is the Ethernet broadcast MAC.

Ethernet II frame: Destination MAC (6 B) | Source MAC (6 B) | EtherType (2 B) | Payload (normally 46–1500 B) | FCS (4 B). Preamble and start-frame delimiter are sent before the frame and often shown separately.

```
show mac address-table
show mac address-table dynamic
show interfaces status
```

> **EXAM ALERT:** A switch learns from the SOURCE MAC address of incoming frames, not the destination MAC.

Practice MCQs

1. A standard Ethernet MAC address is:
A. 16 bits
B. 32 bits
C. 48 bits
D. 128 bits

> **Answer: C** — Six bytes equals 48 bits.

2. Which sublayer handles MAC addressing?
A. LLC
B. MAC
C. Session
D. Transport

> **Answer: B** — MAC is Media Access Control.

6. Network layer and IP addressing

The Network layer provides logical addressing, routing and packet forwarding between networks. A router checks the destination IP, consults its routing table, selects the best matching route and forwards the packet to an outgoing interface or next hop.

6.1 IPv4, network, host and broadcast

IPv4 is 32 bits, written as four decimal octets (0–255). A prefix such as /24 identifies the network bits. The network address has all host bits set to 0; the directed broadcast has all host bits set to 1. In a traditional subnet, usable hosts lie between these addresses.

Example: 192.168.1.0/24 → network 192.168.1.0; usable hosts 192.168.1.1–192.168.1.254; broadcast 192.168.1.255.

6.2 Historical IPv4 classes

Class	First octet	Default mask	Purpose
A	1–126*	255.0.0.0 (/8)	Large unicast networks
B	128–191	255.255.0.0 (/16)	Medium networks
C	192–223	255.255.255.0 (/24)	Smaller networks
D	224–239	N/A	Multicast
E	240–255	N/A	Reserved/experimental

*0 and 127 are special; 127.0.0.0/8 is loopback. Classful addressing is historical; modern networks use CIDR prefixes.

6.3 Private and special IPv4 ranges

Private block	CIDR
10.0.0.0–10.255.255.255	10.0.0.0/8
172.16.0.0–172.31.255.255	172.16.0.0/12
192.168.0.0–192.168.255.255	192.168.0.0/16
Loopback	127.0.0.0/8
Link-local/APIPA	169.254.0.0/16
Limited broadcast	255.255.255.255

Private IPv4 addresses are not globally routed on the public Internet; NAT is commonly used for Internet access. Static addressing is manually configured or reserved. Dynamic addressing is commonly assigned through DHCP.

6.4 IPv6

IPv6 uses 128-bit hexadecimal addresses separated by colons, e.g. 2001:db8:1::10. Leading zeroes in a group may be omitted; one continuous run of zero groups may be compressed once using ::. IPv6 has no broadcast; it uses multicast and anycast. Documentation prefix: 2001:db8::/32.

6.5 Packet forwarding and routing terms

Term	Meaning
Route	Information for reaching a destination prefix.
Next hop	Adjacent router to which packet is sent.
Default route	Fallback route when no more specific match; IPv4 0.0.0.0/0.
Directly connected	Network attached to an active configured interface.
Longest-prefix match	Most specific matching prefix is selected.

A router decrements IPv4 TTL (or IPv6 Hop Limit). IPv4 header checksum is updated. The router removes the incoming Layer 2 frame and creates a new frame for the next link. The IP source/destination normally remain end-to-end unless NAT changes them.

Practice MCQs

1. IPv4 address length?
A. 16
B. 32
C. 48
D. 128

Answer: B — IPv4 is 32 bits.

2. Which is a private IPv4 address?
A. 8.8.8.8
B. 172.20.1.1
C. 224.0.0.1
D. 1.1.1.1

Answer: B — 172.16.0.0/12 is private.

3. Which address class is multicast?
A. A
B. B
C. C
D. D

Answer: D — Class D is 224–239.

4. A router uses which route-selection principle?
A. Longest-prefix match
B. Highest MAC
C. Lowest TCP port
D. Earliest packet

Answer: A — Most specific matching prefix wins.

7. Subnetting: FLSM and VLSM

Subnetting divides a larger IP network into smaller logical networks. It improves address use, limits broadcast domains and supports network segmentation.

7.1 Formulas

Formula	Meaning
Host bits = 32 – prefix	IPv4 has 32 bits.
Total addresses = 2^(host bits)	All combinations of host bits.
Traditional usable hosts = 2^(host bits) – 2	Subtract network and broadcast.
Subnets = 2^(borrowed bits)	Number of equal subdivisions.
Block size = 256 – interesting mask octet	Increment between subnet network IDs.
Wildcard mask = 255.255.255.255 – subnet mask	Inverse of subnet mask.

EXAM ALERT: The normal –2 host rule has exceptions: /31 is used on point-to-point links (two usable addresses under RFC 3021); /32 is a single-host route.

Prefix	Mask	Total addresses	Traditional usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2
/31	255.255.255.254	2	2 point-to-point
/32	255.255.255.255	1	Host route

7.2 Block-size example

Find the subnet containing 192.168.10.77/26. /26 mask = 255.255.255.192. Block size = 256–192 = 64. Ranges are 0–63, 64–127, 128–191 and 192–255. Therefore:

Item	Answer
Network	192.168.10.64
Broadcast	192.168.10.127
Usable range	192.168.10.65–192.168.10.126
Total / usable hosts	64 / 62

7.3 FLSM

Fixed Length Subnet Masking divides a network into equal-sized subnets. Every subnet has the same prefix. Example: split 192.168.1.0/24 into four equal subnets: borrow 2 bits (2^2=4), giving /26. Subnets are .0, .64, .128 and .192; each has 64 total addresses and 62 traditional usable hosts.

7.4 VLSM

Variable Length Subnet Masking permits different subnet sizes. Allocate the largest host requirement first, then the next largest, choosing the smallest block that fits each requirement.

Need	Prefix	Total	Usable	Example allocation
100 hosts	/25	128	126	192.168.10.0/25

Need	Prefix	Total	Usable	Example allocation
50 hosts	/26	64	62	192.168.10.128/26
20 hosts	/27	32	30	192.168.10.192/27
10 hosts	/28	16	14	192.168.10.224/28

The above allocations fit within 192.168.10.0/24. Unallocated addresses remain 192.168.10.240–255.

Feature	FLSM	VLSM
Subnet size	Equal	Different
Mask	Same across subnets	Different prefixes possible
Efficiency	Can waste addresses	More efficient for varied needs
Planning	Simpler	Needs careful alignment

Practice MCQs

1. How many conventional usable hosts in /27?
A. 14
B. 30
C. 32
D. 62

Answer: B — Five host bits: 2^5–2 = 30.

2. Block size for 255.255.255.224?
A. 16
B. 32
C. 64
D. 128

Answer: B — 256–224 = 32.

3. Which method allows different subnet sizes?
A. FLSM
B. VLSM
C. PAT
D. STP

Answer: B — VLSM uses different prefix lengths.

4. In VLSM, allocate which subnet first?
A. Smallest
B. Largest
C. Random
D. Loopback

Answer: B — Largest first reduces address fragmentation.

8. Transport layer: TCP, UDP and ICMP

The Transport layer provides process-to-process communication using port numbers. It can segment data, multiplex application traffic and provide reliability/flow control.

8.1 TCP

Transmission Control Protocol is connection-oriented and provides a reliable, ordered byte stream. It uses sequence numbers, acknowledgements, retransmissions, checksums, flow control and congestion control.

```
TCP three-way handshake:
Client -- SYN --------> Server
Client <-- SYN + ACK -- Server
Client -- ACK --------> Server
Connection established
```

TCP header field	Purpose
Source/destination port	Identify application endpoints.
Sequence number	Numbers bytes in stream.
Acknowledgment number	Indicates next byte expected.

TCP header field	Purpose
Flags	SYN, ACK, FIN, RST, PSH, URG; ECN-related flags also exist.
Window	Receiver-advertised flow-control window.
Checksum	Error detection for TCP header/data.

Flow control protects the receiver using its advertised window. **Congestion control** adapts sending rate to network congestion using a congestion window and algorithms.

8.2 UDP

User Datagram Protocol is connectionless and message-oriented. It has an 8-byte header and no built-in delivery acknowledgement, retransmission, ordering or TCP-style connection setup. Applications may add reliability themselves. It is useful for DNS, DHCP, real-time voice/video and gaming. UDP has a checksum; IPv4 may use zero checksum, while IPv6 generally requires it.

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Built-in ACK/retransmission	No built-in delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Header	Minimum 20 bytes	8 bytes
Typical uses	Web, file transfer, email	DNS, voice/video, DHCP

8.3 ICMP

ICMP is an IP companion/control protocol used for error reporting and diagnostics. Ping uses Echo Request and Echo Reply. Traceroute commonly uses TTL/Hop Limit expiry. Other messages include Destination Unreachable and Time Exceeded. ICMP does not use TCP/UDP port numbers.

> **EXAM ALERT:** ICMP is technically associated with the Internet/Network layer, even though one syllabus row places it under Transport.

Practice MCQs

1. TCP setup normally uses:
A. Two-way handshake
B. Three-way handshake
C. DORA
D. ARP

Answer: B — SYN, SYN-ACK, ACK.

2. Which protocol has an 8-byte header?
A. TCP
B. UDP
C. IPv4
D. Ethernet

Answer: B — UDP header is 8 bytes.

3. Ping commonly uses:
A. ICMP
B. FTP
C. SMTP
D. TCP port 80

Answer: A — Ping uses ICMP Echo messages.

9. Application protocols

Protocol	Purpose	Port(s)
FTP	Traditional file transfer; control TCP 21, active data commonly TCP 20.	TCP 20/21
TFTP	Simple file transfer; initial request uses UDP 69.	UDP 69
SSH	Secure remote terminal.	TCP 22
Telnet	Unencrypted remote terminal.	TCP 23
SMTP	Sends/relays email.	TCP 25; submission commonly 587 or 465
DNS	Name resolution; UDP and TCP.	53
DHCP	Automatic IP configuration; server/client.	UDP 67/68
HTTP	Web request/response.	TCP 80 traditionally
HTTPS	HTTP protected by TLS.	TCP 443 for HTTP/1.1/2; UDP 443 for HTTP/3/QUIC
POP3	Email retrieval/download.	TCP 110; TLS 995
IMAP	Mailbox access and synchronization.	TCP 143; TLS 993

9.1 FTP and TFTP

FTP transfers files between client and server. Traditional FTP is not encrypted. Active mode commonly uses server TCP 20 for data; passive mode tells the client a server-side data port to connect to. SFTP is a separate SSH-based file-transfer protocol, usually TCP 22; FTPS is FTP protected by TLS. TFTP is simpler, uses UDP, and has no built-in authentication/encryption.

9.2 HTTP and HTTPS

HTTP is an application-layer request/response protocol. HTTPS is HTTP protected by TLS, providing encryption, integrity and server authentication. Common methods: GET retrieves; POST submits/processes; PUT creates/replaces; PATCH partially modifies; DELETE requests deletion; HEAD requests headers without a body; OPTIONS describes supported options.

Status	Meaning
200	OK
201	Created
204	No Content
301 / 308	Redirect
400	Bad Request
401	Authentication required/failed
403	Forbidden
404	Not Found
500	Internal Server Error
502	Bad Gateway
503	Service Unavailable

9.3 Email protocols

SMTP sends and relays email. POP3 retrieves/downloads messages. IMAP accesses and synchronizes server-side mailboxes, folders and message state, making it useful across multiple devices.

9.4 DNS

DNS is a distributed hierarchical naming system that resolves domain names to resource records. A recursive resolver may consult root, TLD and authoritative servers; responses are cached according to TTL.

Record	Purpose
A	Name to IPv4 address
AAAA	Name to IPv6 address
CNAME	Alias to canonical name
MX	Mail exchanger and priority
NS	Authoritative name server
PTR	Reverse lookup address to name
TXT	Text data, verification and policy
SOA	Zone metadata and timers
SRV	Service host and port

DNS normally uses UDP 53; TCP 53 is also used, including zone transfers and responses requiring TCP. It is not UDP-only.

9.5 DHCP and DORA

```
Discover → Offer → Request → Acknowledge
Client Server Client Server
```

• Discover: client broadcasts to find DHCP server.

• Offer: server proposes an address and options.

• Request: client requests the offered lease.

• Acknowledge: server confirms lease and configuration.

DHCP uses UDP 67 on server and UDP 68 on client. A DHCP relay forwards requests across subnets when the DHCP server is not on the local broadcast domain.

Practice MCQs

1. Which protocol sends outgoing email?
A. POP3
B. IMAP
C. SMTP
D. DNS

Answer: C — SMTP sends/relays email.

2. Which DNS record stores IPv6 address?
A. A
B. AAAA
C. MX
D. PTR

Answer: B — AAAA is for IPv6.

3. Correct DHCP sequence?
A. DORA
B. RADO
C. ADOR
D. ORDA

Answer: A — Discover, Offer, Request, Acknowledge.

4. Which protocol synchronizes mailbox folders across devices?
A. SMTP
B. IMAP
C. TFTP
D. FTP

Answer: B — IMAP keeps mailbox state on the server.

10. Cisco Packet Tracer and IOS administration

Cisco Packet Tracer is a network simulation/learning tool for creating topologies, configuring routers/switches, connecting end devices and testing simulated connectivity. It does not reproduce every feature or timing detail of physical hardware.

IOS mode	Prompt	Purpose
User EXEC	R1>	Basic monitoring.
Privileged EXEC	R1#	Administrative show/debug commands; enter with enable.
Global configuration	R1(config)#	Global settings; configure terminal.
Interface configuration	R1(config-if)#	Configure interface.
Line configuration	R1(config-line)#	Configure console/VTY lines.
Router configuration	R1(config-router)#	Configure routing process.

Use **end** or Ctrl+Z to return to privileged EXEC; **exit** moves back one level. The question mark (?) provides context-sensitive help; Tab completes commands.

```
enable
configure terminal
hostname R1
enable secret StrongSecret
line console 0
password ConsolePass
login
exit
line vty 0 4
password RemotePass
login
exit
service password-encryption
banner motd #Authorized users only#
end
copy running-config startup-config
```

Enable secret is preferred over legacy enable password. **service password-encryption** obfuscates certain line passwords; it is not strong modern encryption.

Configuration	Meaning	Storage
running-config	Active configuration	RAM (volatile)
startup-config	Loaded at boot	NVRAM on common Cisco platforms

```
show version
show running-config
show startup-config
show ip interface brief
show interfaces
show ip route
show cdp neighbors
show mac address-table
show vlan brief
```

Practice MCQs

1. Privileged EXEC prompt?
A. R1>
B. R1#
C. R1(config)#
D. R1(config-if)#

> **Answer: B** — The hash prompt indicates privileged EXEC.

2. Command to enter global configuration?
A. enable
B. configure terminal
C. show version
D. reload

> **Answer: B** — Run from privileged EXEC.

3. Running-config is normally stored in:
A. RAM
B. ROM
C. Flash only
D. DNS

Answer: A — Running configuration is active in RAM.

11. Remote access: Telnet and SSH

Feature	Telnet	SSH
Purpose	Remote terminal	Secure remote terminal
Transport	TCP	TCP
Default port	23	22
Encryption	None; clear text	Encrypted
Recommendation	Avoid on untrusted networks	Preferred

```
hostname R1
ip domain-name example.local
username admin secret StrongPassword
crypto key generate rsa
ip ssh version 2
line vty 0 4
login local
transport input ssh
```

SSH configuration generally needs a hostname, domain name, local user/AAA and RSA keys. Exact key-size prompts and syntax vary by IOS. Telnet can be enabled on VTY lines but should not be used for secure administration.

Practice MCQs

1. SSH default port?
A. 21
B. 22
C. 23
D. 25

Answer: B — SSH uses TCP 22.

2. Telnet default port?
A. 22
B. 23
C. 53
D. 69

Answer: B — Telnet uses TCP 23.

12. Routing: static, RIP, EIGRP, OSPF and redistribution

Routing selects paths for IP packets between networks. Routes may be directly connected, manually configured (static) or learned dynamically.

12.1 Static routing

Static routes are manually entered and do not automatically adapt to topology changes unless tracking or other mechanisms are used. They are predictable and low-overhead, useful for small/stub networks, but require manual maintenance.

```
ip route <destination-network> <subnet-mask> <next-hop | exit-interface>
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

A floating static route is a backup route configured with a higher administrative distance than the preferred route.

12.2 RIP

Feature	RIP fact
Type	Distance-vector IGP
Metric	Hop count
Maximum usable hops	15; 16 means unreachable
Update	Traditionally 30 seconds
RIPv1	Classful; no subnet mask in updates; no VLSM.
RIPv2	Classless; supports VLSM/CIDR and authentication options.
Cisco AD	120
Transport	UDP port 520

```
router rip
version 2
no auto-summary
network 192.168.1.0
```

RIP is simple but limited and slower to converge. Split horizon, route poisoning and hold-down techniques help reduce loops.

12.3 EIGRP

Feature	EIGRP fact
Type	Advanced distance vector (often called hybrid in older material)
Algorithm	DUAL
Metric	Composite; bandwidth and delay are default components
Cisco AD	90 internal; 170 external
IP protocol number	88 (not a TCP/UDP port)

EIGRP neighbours exchange hello packets. DUAL computes loop-free paths. A successor is the best route; a feasible successor is a qualifying loop-free backup. Named and classic configuration modes exist.

```
router eigrp 100
network 192.168.1.0 0.0.0.255
no auto-summary
```

12.4 OSPF

Feature	OSPF fact
Type	Link-state IGP
Algorithm	Dijkstra SPF
Metric	Cost
Cisco AD	110
IP protocol number	89
Hierarchy	Areas; Area 0 is backbone

OSPF routers exchange link-state information, build a link-state database and run SPF. Single-area OSPF labs commonly place all routers in Area 0. The network command uses a wildcard mask (inverse of subnet mask).

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
```

12.5 Routing comparison and AD

Protocol	Type/algorithm	Metric	Cisco AD
Static	Manual	Administrator-selected	1
RIP	Distance vector	Hop count	120
EIGRP	Advanced distance vector / DUAL	Composite	90 internal / 170 external
OSPF	Link state / SPF	Cost	110

Administrative distance compares trustworthiness of route sources; lower is preferred. Metric chooses the best path within a routing protocol. Common Cisco ADs: connected 0, static 1, eBGP 20, EIGRP internal 90, OSPF 110, RIP 120, EIGRP external 170, iBGP 200.

12.6 Redistribution

Redistribution imports routes from one routing source into another, e.g. RIP into OSPF or EIGRP. It is configured on a router participating in both domains. Plan filters, seed metrics, route tags and loop prevention carefully. Verify with show ip route and show ip protocols.

> **EXAM ALERT:** RIP uses UDP 520. EIGRP uses IP protocol number 88 and OSPF 89; neither uses TCP/UDP port numbers.

Practice MCQs

1. RIP metric?
A. Cost
B. Hop count
C. Delay only
D. Bandwidth only

Answer: B — RIP counts hops; 16 is unreachable.

2. OSPF algorithm?
A. DUAL
B. Dijkstra SPF
C. DORA
D. STP

Answer: B — OSPF is link-state and uses SPF.

3. EIGRP algorithm?
A. DUAL
B. RSA
C. CSMA/CD
D. DORA

Answer: A — EIGRP uses DUAL.

4. Which AD is preferred when same prefix competes?
A. RIP 120
B. OSPF 110
C. EIGRP internal 90
D. EIGRP external 170

Answer: C — Lower AD wins.

13. Backup, restore, TFTP and password recovery

Back up startup configuration, running configuration when needed, and IOS image files. Startup configuration is stored in NVRAM on common platforms; IOS image is commonly in flash. Exact storage and boot procedures vary by device.

TFTP is a simple UDP-based file-transfer service. Its initial request uses UDP port 69; subsequent transfer uses negotiated transfer identifiers/ports. It has no built-in authentication or encryption, so use only in trusted/controlled management networks.

```
copy running-config tftp:
copy startup-config tftp:
copy tftp: startup-config
copy flash: tftp:
copy tftp: flash:
show ip interface brief
```

The device prompts for server address and filename. Confirm reachability, TFTP service, permissions and flash capacity. Password recovery is an authorized console/ROMMON recovery process. The general concept is interrupt boot, enter recovery mode, bypass/ignore saved configuration as supported, load configuration, reset credentials, restore normal boot settings and

save. The exact procedure is model/IOS specific.

> **EXAM ALERT:** Do not memorize a universal configuration-register value or one recovery sequence for all Cisco devices. Follow the platform-specific official recovery procedure.

Practice MCQs

1. TFTP initial request uses:
A. TCP 21
B. UDP 69
C. TCP 22
D. UDP 161

Answer: B — TFTP uses UDP 69 for initial request.

2. IOS image is commonly stored in:
A. Flash
B. ARP cache
C. VTY
D. DNS

Answer: A — Flash commonly stores IOS image files.

14. Security: ACL, NAC, firewall and proxy

14.1 Security principles

Term	Meaning
Confidentiality	Only authorized parties can read information.
Integrity	Information is not improperly altered.
Availability	Systems/data remain accessible.
Authentication	Proving identity.
Authorization	Determining permitted actions.
Accounting	Recording activity for accountability.

14.2 ACL

An Access Control List is an ordered list of permit/deny entries. Cisco evaluates entries top-to-bottom; the first match applies. An implicit deny exists at the end.

Standard ACL: primarily filters on source IPv4 address. Traditional numbered ranges: 1–99 and 1300–1999. **Extended ACL:** can match source/destination IP, protocol and TCP/UDP ports. Traditional numbered ranges: 100–199 and 2000–2699. Named ACLs use descriptive names; either standard or extended ACL may be named.

```
access-list 10 permit 192.168.1.0 0.0.0.255
access-list 101 permit tcp 192.168.1.0 0.0.0.255 any eq 80
interface g0/0
ip access-group 10 in
```

Wildcard mask is inverse of subnet mask: /24 mask 255.255.255.0 → wildcard 0.0.0.255. In a wildcard, 0 means match the bit; 1 means ignore it. **host 192.168.1.10** means one host; **any** means all.

• Standard ACLs are traditionally placed near destination because they only match source address.

• Extended ACLs are traditionally placed near source to stop unwanted traffic early.

• Apply an ACL to an interface and direction (in/out); an un-applied ACL does not filter that interface.

• Put specific rules before broad rules; first match wins; implicit deny at end.

14.3 NAC

Network Access Control checks identity and/or device posture before allowing network access. It may check user identity, device type, patch level and endpoint protection. Non-compliant devices may be quarantined or placed in restricted VLANs.

14.4 Firewalls

Type	Function
Stateless packet filter	Filters individual packets using IP/protocol/port rules.
Stateful inspection	Tracks connection state and permits valid return traffic.
Proxy/application firewall	Intermediary that inspects application requests.
NGFW	May add application awareness, IPS and advanced threat inspection.
Host-based	Runs on endpoint/server.
Network firewall	Protects network boundary/segment.

14.5 Proxies

Type	Role / use
Forward proxy	Client-side intermediary; web filtering, egress control, caching.
Reverse proxy	Server-side intermediary; TLS termination, load balancing, web protection.
Transparent proxy	Intercepts traffic with little client configuration.
Caching proxy	Stores reusable responses to reduce bandwidth/latency.

A proxy is not automatically a firewall, and a firewall is not automatically a proxy. One product may combine functions.

Practice MCQs

1. Standard ACL primarily matches:
A. Source IP
B. Destination port only
C. URL only
D. MAC vendor

Answer: A — Standard ACL matches source IPv4.

2. Implicit final ACL rule:
A. Permit any
B. Deny any
C. Permit TCP
D. Permit ICMP

Answer: B — Unmatched traffic is denied.

3. A forward proxy acts for:
A. Clients
B. Servers
C. Switches
D. DNS roots

Answer: A — Forward proxy represents clients.

4. NAC checks:
A. Device/user compliance
B. Cable colour
C. DNS TTL only
D. TCP window

Answer: A — NAC enforces access posture/policy.

15. NAT: static, dynamic, DNAT and PAT

Network Address Translation modifies IP addressing information as packets pass through a router/firewall. It commonly allows private IPv4 hosts to communicate externally.

Term	Meaning
Inside local	Inside host address as seen internally, commonly private.
Inside global	Address representing inside host to outside, commonly public.
Outside global	Real address of external host.

Term	Meaning
Outside local	Address of outside host as seen from inside; often same as outside global.

15.1 NAT types

Type	Mapping / purpose
Static NAT	Fixed one-to-one mapping; useful for consistently reachable server.
Dynamic NAT	Temporary one-to-one mapping from a public address pool; pool exhaustion blocks new translations.
PAT / NAT overload	Many inside hosts share one public address using transport ports/identifiers.
DNAT	Changes destination address and sometimes port; common for port forwarding to internal server.

```
Static:
ip nat inside source static 192.168.1.10 203.0.113.10
Dynamic:
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask 255.255.255.0
ip nat inside source list 1 pool PUBLIC
PAT:
ip nat inside source list 1 interface g0/1 overload
Verify:
show ip nat translations
show ip nat statistics
```

Interfaces must be identified as **ip nat inside** or **ip nat outside**. NAT is not the same as a firewall: NAT translates addresses; firewall enforces policy. NAT alone is not complete security.

Practice MCQs

1. Fixed one-to-one mapping is:
A. PAT
B. Static NAT
C. Dynamic NAT
D. VLSM

> **Answer: B** — Static NAT is a fixed mapping.

2. Many private hosts sharing one public IP is:
A. Static NAT
B. PAT
C. FLSM
D. STP

> **Answer: B** — PAT differentiates sessions using ports.

3. Port forwarding to internal service is commonly:
A. DNAT
B. RIP
C. VTP
D. FLSM

> **Answer: A** — DNAT changes destination address/port.

16. Switching: VLAN, DTP, VTP, STP, Port Security

16.1 VLAN

A VLAN logically divides a switched network into separate Layer 2 broadcast domains. Devices in different VLANs normally require Layer 3 routing to communicate. VLANs improve segmentation and administration but are not a substitute for security controls.

Port type	Traffic
Access	Normally one assigned VLAN; typical endpoint port.
Trunk	Multiple VLANs over one link; typical switch-to-switch or switch-to-router.

Port type	Traffic
802.1Q	IEEE VLAN tagging; native VLAN traffic is commonly untagged.

```
show vlan brief
vlan 10
name SALES
interface fa0/1
switchport mode access
switchport access vlan 10
interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
switchport trunk native vlan 99
```

16.2 DTP

Dynamic Trunking Protocol is Cisco-proprietary and negotiates trunk formation on compatible switches. Modes: access (force access), trunk (force trunk), dynamic desirable (actively tries), dynamic auto (passively waits), nonegotiate (disable negotiation on supported trunk). Desirable + auto forms trunk; auto + auto generally does not. Explicit configuration is more predictable.

16.3 VTP

VLAN Trunking Protocol is Cisco-proprietary and distributes VLAN database information in a VTP domain over trunk links. Server can create/modify/delete VLANs; client receives database changes; transparent maintains its local VLAN database independently and may forward advertisements depending on version. Carefully manage domain, password, version and revision number; a higher revision may overwrite VLAN information in some designs.

16.4 STP

Spanning Tree Protocol prevents Layer 2 loops caused by redundant links. Without STP, loops can create broadcast storms, duplicate frames and MAC-table instability. STP elects the switch with the lowest Bridge ID as root. Non-root switches select root ports; segments select designated ports; redundant ports are blocked/discarding.

Term	Meaning
Root bridge	Switch with lowest Bridge ID.
Root port	Best path toward root on a non-root switch.
Designated port	Forwarding port for a LAN segment.
BPDU	Bridge Protocol Data Unit exchanged by STP switches.
Classic 802.1D states	Blocking, Listening, Learning, Forwarding, Disabled.
RSTP states	Discarding, Learning, Forwarding; generally faster convergence.

16.5 Port security

Port security restricts source MAC addresses allowed on a switch access port. It can limit MAC count and use static, dynamic or sticky secure MAC addresses.

Violation mode	Action
Protect	Drops frames from unauthorized MACs; limited/no notification in classic behaviour.
Restrict	Drops frames, increments violation counter and may notify.
Shutdown	Places port into err-disabled state; default on many Cisco switches.

```
interface fa0/1
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
show port-security interface fa0/1
```

16.6 Inter-VLAN routing

Router-on-a-Stick (ROS): one physical router interface has multiple subinterfaces, each tagged for a VLAN using 802.1Q. The switch-to-router link is a trunk.

```
interface g0/0
no shutdown
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

SVI: a Switched Virtual Interface is a virtual Layer 3 interface associated with a VLAN on a multilayer switch. It can act as default gateway. Inter-VLAN routing requires Layer 3 routing enabled.

```
ip routing
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
```

An SVI typically comes up only when the VLAN exists and an active Layer 2 port or trunk carries it, subject to platform conditions.

Technology	Purpose
VLAN	Layer 2 broadcast segmentation
DTP	Trunk negotiation
VTP	VLAN database distribution
STP	Loop prevention
Port security	Limits allowed MAC addresses
ROS	Router subinterfaces over trunk
SVI	Virtual Layer 3 interface for VLAN

Practice MCQs

1. VLAN creates a separate:
A. Broadcast domain
B. TCP session
C. DNS zone
D. Internet

Answer: A — Each VLAN is a separate Layer 2 broadcast domain.

2. IEEE trunk tagging standard:
A. 802.1Q
B. 802.11
C. 802.3 only
D. 802.1X

Answer: A — 802.1Q tags VLAN frames.

3. Which protocol prevents Layer 2 loops?
A. VTP
B. DTP
C. STP
D. DHCP

Answer: C — STP creates a loop-free logical topology.

4. Root bridge is selected using:
A. Lowest Bridge ID
B. Highest IP
C. Highest port count
D. First powered on always

Answer: A — Lowest Bridge ID wins.

5. Port-security shutdown violation causes:
A. VLAN deletion
B. Err-disabled port

C. DNS flush
D. NAT pool exhaustion

Answer: B — Shutdown mode disables the port.

17. Web services and practical lab coverage

A web service exposes functionality/data over a network using defined interfaces and message formats. A client sends an HTTP/HTTPS request; server validates and processes it; server returns status, headers and optional body. SOAP uses XML-based messages; REST is an architectural style commonly implemented using HTTP methods and resource URLs. JSON is common but REST is not limited to JSON.

Practical lab	Core task and verification
1.1 CLI modes/components	Identify router/switch components and navigate CLI modes.
1.2 Help	Use ? and Tab in different modes.
1.3 Hostname/password/banner	Configure hostname, enable secret, console/VTY and MOTD.
1.4 Save config	Copy running-config to startup-config; verify.
2 Connectivity/cables	Connect devices, assign IPs and test ping.
3 MAC table	Generate traffic; use show mac address-table.
4.1 IP schema	Plan subnets and calculate host ranges.
4.2 Physical/loopback IP	Configure addresses and verify interface status.
TCP/UDP header demo	Compare ports, flags, sequence/ack and UDP header.
HTTP/HTTPS flow	Use simulation mode to inspect packet exchange.
FTP/DHCP/DNS/SMTP/POP3	Configure services and test clients.
Telnet/SSH	Configure remote access; explain SSH security.
Static/RIP/OSPF/EIGRP	Configure routes and verify end-to-end reachability.
Redistribution	Exchange selected routes between routing domains.
TFTP backup/restore	Copy configuration to/from TFTP server.
Password recovery	Understand platform-specific authorized console recovery.
Standard/Extended ACL	Create named/numbered ACLs, apply direction and test.
Static/Dynamic NAT/PAT	Configure translations and inspect NAT table.
VLAN management	Create, rename, remove VLANs and assign access ports.
DTP/VTP	Configure trunk negotiation and controlled VLAN database.
Port security	Limit MACs and test violation actions.
ROS/SVI	Configure gateways and test inter-VLAN communication.

Troubleshooting sequence

• Check physical link, power, cable and interface status.

• Check IP, mask, gateway and DNS settings.

• Ping local interface/gateway, remote gateway, then remote host.

• Check VLAN assignment, trunk allowed VLANs and native VLAN.

• Check routing table and next-hop reachability.

• Check ACL/firewall rules and NAT translations.

• Use show commands and Packet Tracer Simulation mode; change one thing at a time.

```
ping <ip>
traceroute <destination> (Cisco IOS)
tracert <destination> (Windows)
```

```
ipconfig /all (Windows)
show ip interface brief
show interfaces status
show vlan brief
show interfaces trunk
show mac address-table
show ip route
show ip protocols
show access-lists
show ip nat translations
show spanning-tree
```

Practice MCQs

1. HTTP 404 means:
A. OK
B. Redirect
C. Not Found
D. Server Error

> **Answer: C** — 404 indicates resource not found.

2. Which is commonly used as REST data format?
A. JSON
B. VGA
C. STP
D. ARP

> **Answer: A** — JSON is common; REST is not restricted to it.

18. Final revision sheets

18.1 OSI quick recall

Layer	Name	Keyword
7	Application	HTTP, FTP, DNS
6	Presentation	Format, encryption concept
5	Session	Dialog/session
4	Transport	TCP/UDP, ports
3	Network	IP, routing
2	Data Link	Ethernet, MAC, frames
1	Physical	Bits, signals, cables

18.2 Ports and protocol facts

Port/protocol	Remember
20/21 TCP FTP	Active data commonly 20; control 21.
22 TCP SSH	Secure remote access.
23 TCP Telnet	Unencrypted remote terminal.
25 TCP SMTP	Mail relay.
53 UDP/TCP DNS	Name resolution; TCP also used.
67/68 UDP DHCP	Server/client.
69 UDP TFTP	Initial request.
80 TCP HTTP	Traditional web.
110 TCP POP3	Email retrieval.
143 TCP IMAP	Email synchronization/access.
443 TCP/UDP HTTPS	TCP for HTTP/1.1/2; QUIC/HTTP/3 uses UDP.
520 UDP RIP	RIP routing updates.
88 EIGRP / 89 OSPF	IP protocol numbers, not ports.

18.3 Essential MCQ traps

• Hub = Layer 1; switch = Layer 2; router = Layer 3.

• Switch learns SOURCE MAC, then looks up destination MAC.

• Switch port is one collision domain; VLANs separate broadcast domains.

• Router separates broadcast domains by default.

• Standard ACL matches source IPv4; extended ACL can match source/destination/protocol/ports.

• ACL is top-down; first match wins; implicit deny at end.

• Telnet TCP 23 is clear text; SSH TCP 22 is encrypted.

• RIP hop count 16 means unreachable; maximum usable is 15.

• OSPF uses SPF; EIGRP uses DUAL; RIP uses hop count.

• RIP uses UDP 520; OSPF protocol 89; EIGRP protocol 88.

• VLAN segmentation; DTP trunk negotiation; VTP VLAN database distribution; STP loop prevention.

• PAT allows multiple private hosts to share one public address using ports.

• ICMP is a Network/Internet-layer control protocol.

• DNS uses UDP and TCP 53; SMTP sends; POP3/IMAP retrieve/access.

• DHCP DORA = Discover, Offer, Request, Acknowledge.

• HTTPS uses TLS; HTTP/3 uses QUIC over UDP.

18.4 Exam time strategy

• 60 questions in 120 minutes gives an average of 2 minutes per question.

• First pass: answer confident questions; mark uncertain ones.

• Second pass: solve subnetting and revisit tricky comparisons.

• For subnetting, write prefix, block size, network and broadcast before selecting an option.

• Watch NOT, EXCEPT, FIRST, BEST, ALWAYS and ONLY.

• If no negative marking, attempt all questions; if negative marking exists, follow the scoring rule.

EXAM ALERT: Highest-return revision: OSI layers, ports, subnetting, TCP vs UDP, routing metrics/ADs, ACL rules, NAT types, VLAN/DTP/VTP/STP and Cisco CLI modes.

Mixed mock test — 20 questions

Practice MCQs

1. Which layer creates frames?
A. Physical
B. Data Link
C. Network
D. Transport

Answer: B — Data Link encapsulates packets into frames.

2. Which protocol resolves an IPv4 next-hop address to a local MAC?
A. DNS
B. ARP
C. DHCP
D. SMTP

Answer: B — ARP resolves IPv4-to-link-layer address on the local link.

3. Which port is SSH?
A. 21
B. 22
C. 23
D. 25

Answer: B — TCP 22.

4. Which routing protocol is link-state?
A. RIP
B. OSPF
C. FTP
D. DHCP

Answer: B — OSPF is link-state.

5. RIP maximum usable hop count?
A. 14
B. 15
C. 16
D. 255

Answer: B — 16 is unreachable.

6. Which ACL type can filter destination TCP port?
A. Standard
B. Extended
C. Neither
D. Only MAC ACL

Answer: B — Extended ACL can match protocol and ports.

7. Which record identifies mail exchanger?
A. A
B. AAAA
C. MX
D. PTR

Answer: C — MX identifies mail servers.

8. A /28 provides how many traditional usable hosts?
A. 8
B. 14
C. 16
D. 30

Answer: B — Four host bits: 16 total minus network/broadcast.

9. Which address is limited IPv4 broadcast?
A. 0.0.0.0
B. 127.0.0.1
C. 255.255.255.255
D. 224.0.0.1

Answer: C — Limited broadcast.

10. What does TTL help prevent?
A. Indefinite packet looping
B. MAC learning
C. Encryption failure
D. DHCP lease

Answer: A — TTL decreases at each router.

11. Which device is a multiport repeater?
A. Router
B. Hub
C. Firewall
D. Load balancer

Answer: B — Hub repeats signals.

12. Which command saves running config?
A. reload
B. copy running-config startup-config
C. erase startup-config
D. show version

Answer: B — Saves active config for next boot.

13. Which technology distributes VLAN database in Cisco networks?
A. DTP
B. VTP
C. STP
D. NAT

Answer: B — VTP distributes VLAN information.

14. Which technology negotiates trunk links?
A. DTP
B. DHCP
C. DNS
D. OSPF

Answer: A — DTP negotiates trunking.

15. Which method shares one public IP among many hosts?
A. Static NAT
B. PAT
C. FLSM
D. STP

Answer: B — PAT uses transport identifiers/ports.

16. Which ICMP message is used for ping response?
A. Echo Reply
B. SYN ACK
C. DHCP Offer
D. DNS MX

Answer: A — Ping uses Echo Request and Echo Reply.

17. Which route wins among matching prefixes?
A. Least specific
B. Longest prefix
C. Oldest
D. Highest TTL

Answer: B — Longest-prefix match.

18. Which statement about IPv6 is correct?
A. 32-bit
B. Uses broadcast
C. 128-bit
D. Dotted decimal

Answer: C — IPv6 uses 128-bit hexadecimal addresses.

19. Which port is TFTP initial request?
A. TCP 69
B. UDP 69
C. TCP 22
D. UDP 53

Answer: B — TFTP uses UDP 69.

20. HTTPS means:
A. HTTP protected by TLS
B. FTP over UDP
C. Unencrypted HTTP
D. Routing protocol

Answer: A — TLS protects the HTTP exchange.